Global Facilities - Design & Construction Standard - CSA - Earthworks Specifications

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1.0 Purpose

1.1 This document provides a Required Standard and Standard Guideline on Material and Design Specification that shall be followed and complied in the Design, Engineering and Construction for Earthworks as part of any green and/or brown field construction projects contract requirement.

1.2 The standard is applicable for any green and/or brown field construction projects as per required in the contract.

1.3 The document is intended to serve as a reference and guideline in addition to any Construction Drawings and Specification provided in the contract requirement including any required by local authority regulation and standard requirement as well as code of practice.

1.4 The established Specifications Standard for each locale shall also comply with Micron’s in-house:

1.4.1 RBA Code of Conduct requirements for Semiconductor Manufacturing

1.4.2 Environmental, Health & Safety Standards

1.4.3 Operational Requirements /SOP (Standard Operation Procedures)

1.4.4 Security Protocol

1.5 The Specifications Standard to be adopted in each local shall not violate any local codes or by-laws and reflects the current semi-conductor industry's best practices.

1.6 Where there is the option to adopt either the MasterSpec or in-region specifications standard, the AE (Architectural & Engineering Consultants) shall propose the most suitable solution based on safety, quality, time & cost to Micron's Designated Project Team (DPT) for review and acceptance. Otherwise, the more stringent option shall apply.

2.0 Scope

2.1 This document for earthworks specification covers all necessary related design, engineering, procurement, construction, installation and completion of piling, substructure, superstructure, and roof structure to full compliance of the contract requirement and required for the project. It extended to cover any green and/or brown field construction projects.

2.2 This document applies to all Micron Facilities employees, contractors, and vendors at all Micron sites.

3.0 Definitions and Acronyms

3.1 Definitions of acronyms used in this document are listed as follows:

Term/Acronym	Definition / Description
FCT	Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKMs
GFTT	Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
GFC	Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
Global Facilities Central Team	This is used interchangeably with Innovation & Integration Team
Global Facilities Construction Team	This is used interchangeably with Global Construction Team
Global Facilities Operations Support Team	This is used interchangeably with Global Facilities Operations Team
Global Facilities Planning Team	This is used interchangeably with Global Facilities Line Planning Team

Term/Acronym	Definition / Description
AE	External Architectural & Engineering Consultants
DPT	Micron's Designated Project Team
GFCT	Micron's Global Facilities Construction Team
EHS	Micron's Environmental Health & Safety Team
QP/PE	External Qualified Person (professional architect/engineer)
SME	Micron's Subject Matter Expert
TM	Micron's Team Members

4.0 General

4.1 Earthworks shall generally comply with the requirements of C2-20 General Earthworks Specifications NPQS or equivalent local codes.

4.2 All earthworks shall be finished to uniform surfaces which shall conform to the following limits for the lines, grades and widths shown in the approved drawing:

4.2.1 Formation Width

The widths measured on earth side from the centre line to the toe of cut and/or the top edge of fill shall not be less than the width specified in the design and no portion of the cut slope shall encroach within the formation width.

4.2.2 Formation Level

The finished surface shall not vary by more than 25mm above or below the specified levels and no points in the general surface shall vary by more than 12mm from a 3 metre straight edge laid parallel to the centre line of the road or from a template placed at right angles to the centre line.

4.2.3 Subgrade Width

The subgrade width shall not be less than the specified width shown in the drawing nor more than 150mm greater than the specified width.

4.2.4 <u>Slopes</u>

At all levels, the average slope shall not be steeper than the slopes specified and no point on the completed slope shall vary from the average plane by more than 150mm in the case of earth slope in cut or 300mm in the case of fill slope.

4.3 The Builder shall at all times keep the site free of standing water by means of temporary drainage. Before the conclusion of each day's work, the Builder shall grade the site to prevent ponding of water should it rain during the night.

5.0 Removal of Vegetation and Topsoil

5.1 The area affected by earthwork shall be grabbed free from heavy grass, vegetation, decayed stumps, roots and other perishable matter. Within the areas where excavation will be made, all stumps, roots and other objectionable material shall be removed to a depth of not less than 300mm below the finished surface of the subgrade.

5.2 All topsoil in areas affected by earthwork including embankments required to receive additional fill shall be excavated and stock pile at an approved location for re-use later. All surplus topsoil shall be removed to approved site after the completion of the works.

Topsoil shall be removed in accordance with BS 3882 or equivalent local code.

5.3 Trees

The Builder shall bring to the attention of the S.O. / Engineer, any trees and / or roots affected by any excavation works. Unless indicated as 'trees to be felled' in the drawings, approval for the cutting of trees and roots must be obtained from the S.O. / Engineer before works commence.

6.0 Excavation

6.1 All excavation work shall include excavation, removal and satisfactory disposal of all materials from within the limits of the works. It shall also include all excavation, shaping and sloping for the construction, preparation and completion of the subgrade, shoulders, batters, drains, intersections, approaches, ramps, etc. to the required alignments, grade and cross section as designed.

6.2 Unless otherwise specified or shown in the approved drawings, maximum slopes to cuttings shall conform to the following **Table 1**:

Table 1

Materials	Maximum Slope (Horizontal Distance to Vertical Rise)
Sand	(3 : 1)
Loose gravel or medium clay, loam	(1½ : 1)
Shale or similar soft rock	(1 : 1)
Jointed laminated or soft rock	(½ : 1)
Massive rock	(¼ : 1)

6.3 When according to the S.O.'s assessment of the type of materials at the cut site, the slopes as proposed on the approved drawings have to be re-determined, the Builder shall cut the slope as instructed and the Builder shall not be entitled to claim for the additional work except when the re-determination is made for purposes other than to ensure the stability of the cutting. If however, the Builder excavates the slope of a cutting beyond the specified line and the tolerance applicable thereto, the Builder may be instructed to fill and make up the cutting to the required slope, in the manner as directed by the S.O., all at the Builder's own expense. Such a change shall not be regarded as a re-determination of the slope and no payment shall be made.

6.4 Excavation for trenches for foundations, drain pipes, etc. shall be carried out to the dimensions, depth levels as indicated in the drawing. If however, the Builder excavates to any greater widths or depths than shown on the drawings for foundations or as instructed by the S.O. then he shall, at his own expense, fill in such depths or widths of excavation with grade 30 concrete.

6.5 The bottoms of all excavations shall be free from mud and water, trimmed clean protected from the effects of weather and thoroughly consolidated to the satisfaction of the S.O. by rollers, rammers or other approved methods before placing any constructional material. All soft or defective portions shall be cut out and filled in with selected excavated material well consolidated in layers not exceeding 150mm thick, except for foundation which is covered in Clause 3.4 above.

6.6 Where required, the Builder shall design and provide all necessary planking and strutting and sheet piles if necessary to uphold the face of the excavation and any necessary staging without additional cost.

7.0 Removal of Rock and Other Blasting Operations

7.1 Should rock be met in the course of excavation, it must be removed by approved means.

7.2 Blasting will not be allowed without written permission from the S.O. If explosives are to be used, the Builder shall obtain all necessary licenses from the appropriate authorities and shall conform to all Government regulations relating to transport storage, handling and use of the explosives and shall also conform to the rules set out by the Officer-in-Charge of Arms and Explosives.

7.3 Rock is material, which in the opinion of the S.O., can only be excavated by the use of wedges or compressed air plant. Rock is to be stacked on site for measurement before being carted away. Rocks removed without prior measurement will not be paid later

7.4 Excavation in solid rock shall mean excavation in rock found in ledges, large bounders or masses in its original position which would normally have to be lessened by pneumatic tools or if excavated by hand, by wedges and sledge hammers. All solid boulders or detached pieces or rock exceeding $0.38m^3$ in volume in trenches or exceeding $0.76m^3$ in general excavation but not otherwise shall be regarded as solid rock.

7.5 Material other than rock including but restricted to earth gravel and also such hard and complex material as shall which can be removed by ordinary excavating machines and also boulders or detached pieces of solid rock not exceeding $0.38m^3$ in volume in trenches or $0.76m^3$ in general excavation shall be regarded as ordinary materials.

7.6 On encountering rock, the Builder shall at once notify the S.O. the existence of such material. The S.O. shall reserve the right to decide whether or not such material is rock and his decision shall be final.

8.0 Removal of Soft Soil

In cuttings where soft and unsuitable materials occur within one metre below the design levels of the subgrade, such materials shall be removed and replaced with approved stable materials in layers not exceeding 150mm loose thickness, compacted as elsewhere specified.

9.0 Approval of Excavation

The Builder shall report to the S.O. when secured bottoms have been obtained in the excavations and are ready to receive the foundations. Any concrete or other work put in before the S.O.'s approval shall, if so directed, be removed and replaced all at the Builder's own expense.

10.0 Disposal of Surplus Excavated Material

Surplus excavated material arising from excavation and not required for filling, etc. shall be removed, deposited, spread, levelled and mechanically rammed and consolidated on site where directed by the S.O. or removed off the site. The Builder shall find his own area for dumping which has to be approved by the S.O.

11.0 Earth Fill

11.1 Approved material shall be used for fill construction including embankments, and shall be free from logs, stumps, weeds and organic matter or any other deleterious matter. The filling material shall consist of suitable material all of which shall pass a125mm BS sieve and at least 95% shall pass the 75mm BS sieve.

The coefficient of uniformity shall be greater than 10. The fraction passing a 75m BS sieve shall be less than 20% by weight and shall have the following characteristics:

a) Liquid limit not exceeding 35:
b) Plasticity index not exceeding 12.

11.2 Areas and embankments on which fills will be constructed shall first be cleared and grubbed as specified. Top soil to full depth (including turf) shall be removed. Filling shall not be placed until the approval of the S.O. has been obtained.

11.3 Embankments shall be constructed in such order and manner that adequate drainage of the working areas is maintained throughout the construction period.

11.4 Surfaces with slopes steeper than one vertical to four horizontal shall be cut into a series of level benches before filling. Filling shall be placed in horizontal layers beginning at the lowest point in the natural surface and shall be constructed to the full dimensions of the embankment at each layer. Materials which have been loosened shall be compacted together with the filling placed in the succeeding layer. The loose thickness including the newly placed fill of the layers shall not exceed 200mm.

11.5 All fills shall be constructed in layers of uniform thickness not exceeding 200mm and each layer shall be compacted as specified in Clause 11. Layers of filling may be formed by equipment which will spread the material as it is dumped, or by blading or spreading by other acceptable methods so that the material is uniformly distributed.

11.6 Before beginning compaction, the material in each layer of the embankment shall be uniform in composition and moisture content. Clods of material shall be broken and the material mixed by blading, harrowing, disking or by other methods. Oversized rocks shall be broken to the specified maximum dimension of 125mm. Rock shall be placed, spread and compacted in such a manner that the interstices between the larger pieces are filled with compacted finer materials.

11.7 Unless otherwise specified or shown on the drawings, the maximum slopes will conform with the following **Table 2**:

Table 2

Material	Maximum Slope (Horizontal Distance to Vertical Rise)
Sandy loam, clay and loose sand	(3 : 1)
Ordinary lateritic earth	(1½ : 1)
Rock	(1¼ : 1)
Rock filling-hand pitched	(1 : 1)

11.8 When the S.O. redetermines a slope, the Builder shall not be entitled to claim whatsoever, unless such redetermination is for the purpose other than to ensure the stability of the slope.

12.0 Fill Adjacent To Bridge Abutments, Wing, and Retaining Walls

12.1 Material adjacent to weepholes in abutments, wing and retaining walls shall consist of clean, hard and durable broken stone, graded from 50mm to 9mm sizes of particles. The larger particles shall be placed adjacent to the weepholes and the small particles behind and above the larger particles.

12.2 The graded broken stone shall extend horizontally at least 300mm from each weephole and at least 450mm vertically above the weephole.

12.3 In addition to the graded broken stone at weepholes, selected fill consisting of granular material having a maximum size of 50mm, a Plasticity Index of not more than twelve (12) and having at least sixty (60) per cent retained on a No. 7 B.S. Sieve (2.40mm) shall be placed adjacent to bridges, culverts and walls in accordance with the following tables **Table 3**:

Table 3

Structure	Minimum Width of Selected Fill
Bridge abutment and wing walls	2.0 m
Culvert wing walls	H/3
Retaining walls	H/3
Barrels of box culverts	H/3
Barrels of pipe culverts	600 mm
(Where H = Height of Structure)	

12.4 The S.O. shall determine whether the material proposed by the Builder is suitable for use as selected fill. If in the opinion of the S.O. selected material of the required quality is unobtainable from the excavation under the Contract, he may give authority for the material to be obtained from other borrowed sites with no extra payment.

12.5 The selected fill shall be placed in layers of 150mm up to the subgrade level. Compaction shall start at the wall and proceed away from it and shall be carried out as specified in Clause 11.

12.6 Where the slope of the natural surface behind abutment walls and wing walls exceeds one (1) vertical to four (4) horizontal, the slope shall be cut in the form of successive horizontal terraces at least 600mm in width.

12.7 No fill shall be placed against abutments or wing walls of concrete structures within 14 days after placing concrete in the abutments or walls and in the superstructure of the adjacent span, unless the walls are properly strutted to the approval of the S.O.

12.8 In the case of spill-through abutments, rock fill shall not be dumped against the columns but shall be built up evenly by hand placing around individual columns.

12.9 In the case of framed structures, fill at both ends of the structures shall be brought up simultaneously, the difference between the levels of the fills at the respective abutments not to exceed 600mm.

13.0 Compaction Plant

13.1 Compaction methods and compaction plants used shall be in accordance with Table 4.7.2 of C2-20 General Earthworks NPQS or equivalent local codes.

The Builder shall provide and operate sufficient compaction plants of suitable type (to be approved by the S.O.) to compact embankments, subgrades and pavement courses in accordance with these specifications.

13.2 Rollers to be provided and their towing tractors if any, shall be in good mechanical condition and shall conform to the following respective requirements:

13.2.1 Smooth wheel rollers and grid rollers shall weigh not less than 10 tones and shall provide compression under the rear wheels of not less than 61N per lin mm of width.

13.2.2 Tamping rollers shall be of more than 40 kN category and shall consist of metal rollers or drums studded with tamping feet with projection not less than 180mm from the face of the drum. The tamping feet shall first have to be arranged in circumferential rows so that the clear spacing between the outer ends of the feet is not less than 150mm nor more than 300mm in the rows and not more than 300mm between the rows. The space between the circumferential rows shall be properly cleaned by spacer bars. The cross sectional area of each tamping foot measured perpendicular to the diameter of the roller shall be not less than 45 sq cm and shall not be more than a minimum load in each tamping foot of 1.7 N per sq mm of cross sectional area. This load per tamping foot shall be determined by dividing the total weight of the roller by the maximum number of the roller by the maximum number of feet in a row parallel to or approximately parallel to the axis of the roller. Rollers with feet which, in the opinion of the S.O., are excessively worn, shall not be used.

13.2.3 Pneumatic Tyred Rollers

Tyre pressures shall not be less than 552 KN/m² and the load per tyre shall not be less than 22240N. Double axle type rollers shall have the rear tyre staggered in relation to the front tyres.

13.2.4 Vibrating smooth drum rollers shall have a total weight of not less than 4 tonnes and shall have a static load on the vibrating roll of 0.25 – 0.45 kN per 100mm width. Also, the ratio of the centrifugal force produced by the rotating mass to the weight of the vibrating roller shall not be less than one.

13.2.5 Vibration tamping rollers shall weigh not less than 4½ tonnes and shall comply with the requirements for tamping rollers in Clause 10.2.2 above except that the cross sectional area of each tamping foot shall be such that the weight of the roller divided by the total cross sectional area of all of the feet on the roller shall be not less than 42.7 kN/m².

13.2.6 Any other type of compacting equipment which the Builder may desire to use shall first have to be approved by the S.O. before being put into service. If it fails to produce the specified compaction, its use shall be discontinued and other approved rollers shall be provided immediately.

14.0 Compaction of Earthworks

14.1 Subgrade in cuttings, areas upon which fills are to be placed and all fill material shall be compacted to the standards indicated hereunder. All compaction shall be carried out using approved mechanical plants in accordance with Table 4.7.2 of C2-20 General Earthworks NPQS.

The depth of each compacted layer shall not be greater than the maximum depth of the compacted layer stipulated for each type of compaction plant.

Earthmoving plant shall not be accepted as compaction equipment.

Alternative methods or plants may be accepted if it can be demonstrated to the satisfaction of the SO at site trials that the required field density can be achieved.

14.2 Before commencement of any filling, each class of fill material to be compacted shall be tested by an accredited laboratory to establish the maximum value of the dry density that can be achieved and the optimum moisture content for compaction. Tests shall be carried out in accordance with BS 1377 test No. 13.

14.3 Each layer of material placed (including sub-base layers), the natural surface in areas to be occupied by fills, the natural surface at the junction of cuts and fills, materials backfilled in cutting and material replacing unstable portions of the natural surface with selected material shall be trimmed as construction proceeds and shall be uniformly compacted to the required density in accordance with this sub-clause before the laying of the next layer is commenced. The compaction of undisturbed natural ground shall be to the required density for a depth of not less than 200mm.

14.4 The in-situ field densities of all compacted materials calculated as a percentage of the maximum dry density derived from BS 1377 Test No. 13 (4.5 kg rammer) shall not be less than:

Within 500mm of formation level - 95 %
Below 500mm of formation level - 90 %

Notwithstanding compliance with Table 4.7.2 in General Earthworks NPQS, the Builder shall carry out site tests at a rate of 1 test per 400 m^2 of surface area of each compacted layer to verify that the required field density is achieved, and make any adjustments to the compaction methods if necessary.

14.5 At the time of compaction of each layer, the moisture content of the material shall be adjusted so as to obtain the degree of compaction specified. When directed by the S.O., water shall be added to material which contains insufficient moisture for compaction. The added water shall be sprayed uniformly and thoroughly mixed with material until a homogenous mixture is obtained. Material containing excessive moisture shall not be compacted until the material has dried out sufficiently to obtain the required compaction.

14.6 Compaction shall be undertaken by any means necessary to obtain the specified compaction for the full depth of each layer in fills and for the full width of the formation over the entire length of the work. At locations where it would be impracticable to use mobile power compacting equipment, fill layers shall be compacted to the specified requirements by any means approved by the S.O. that will obtain the specified compaction.

14.7 Construction equipment and traffic shall not be allowed on the subgrade or fill while it is in a wet condition. Material which has become excessively wet shall be dried or removed from the site and replaced by material of suitable moisture content for compaction at the Builder's expense.

14.8 Filling over and around pipes, culverts, bridges and other structures shall be compacted in such a manner that will avoid unbalanced loading and that will not cause movement or place strain on any structures.

14.9 The Builder shall carry out the following tests to demonstrate that adequate compaction in accordance with the Specifications has been carried out to the subgrade or fill layers :

a) Dry Density - Moisture content of subgrade or fill material.

b) In-situ Field Density Tests.

c) California Bearing Ratio Test

15.0 Drainage of Excavations

15.1 The Builder shall arrange for the rapid dispersal of water shed on to the site from any source. Where practicable, the water shall be discharged into the permanent outfall for the drainage system. Adequate means for trapping silt on temporary systems discharging into permanent drainage systems shall be provided.

15.2 Where necessary, temporary water courses, ditches, drains, pumping or other means of maintaining the earthworks free from water shall be provided. The Builder shall also maintain a sufficient minimum surface crossfall at all times and, where practicable, a sufficient longitudinal gradient to enable them to shed water and prevent ponding.

16.0 Geotextile Sheet

16.1 Unless otherwise specified, geotextiles shall be non-woven and shall be made of polypropylene, polyethylene, polyester or a combination of the aforesaid materials.

Geotextiles shall not be susceptible to bacteria and fungus attack and shall be resistant to chemical action and not affected by exposure to ultra-violet light.

Class A geotextiles shall be used where the geotextiles serve as a separation or reinforcement function. Class B geotextiles shall be used where the geotextiles serve as a filtration function. The property requirements for class A and class B geotextiles are specified below.

The Builder shall submit methods of laying and jointing prior to commencement of work

Table 4

PHYSICAL PROPERTIES OF GEOTEXTILES			
Physical Property	Class of Geotextile		Specification
	Class A	Class B	
Minimum Unit Weight, m/^{2}	125	50	ASTM D5261 or ISO 9864
Minimum Grab Tensile Strength, N	530	270	ASTM D4632
Minimum Elongation To Break, %	60	55	ASTM D4595 or ISO 10319
Minimum Trapezoidal Tear Strength, N	180	80	ASTM D4533
Water Permeability, cm/sec	2.0×10^{-2}	2.7×10^{-2}	Manufacturer's test report to be submitted.

16.2 In the use of geotextiles sheets, the following shall be complied with:

a) Protect from exposure to light, except for a maximum of 5 hours during laying. Protect from all materials listed as potentially deleterious by the manufacturer.

b) Before commencing laying, remove humps and sharp projections from the ground and fill hollows. Do not stretch or bridge the geotextiles over irregularities and ensure that the specified lap is maintained.

c) Take care to avoid damage to geotextile by vehicles, plant or by tipping material from excessive height. Do not allow construction traffic over geotextiles until covered by the full thickness of fill.

d) Temporarily weight edges of sheet with fill and lay a maximum of 15m of geotextiles before covering to prevent uplift. Place fill as soon as possible after laying and within a maximum of 24 hours.

e) Place geotextiles patches over tears with minimum lay of 300mm beyond extent of damage and cover immediately to retain in place.

16.3 In jointing geotextile sheets, the Builder shall propose and comply with one of the following methods of jointing:

a) 300mm overlap – Overlaps to be 300 to 1000mm, with the wider overlap necessary on very soft subgrades to allow for movement when trafficked by vehicles. Overlaps to be weighted down immediately with small amounts of aggregate.

b) Single stitched prayer seam with high tenacity polyester thread – Carry out sewing with portable stitching machines using high tenacity polyester or polyester/ cotton thread in a single or double stitch to form a prayer seam configuration. A single line stitch is generally adequate.

c) Single welt with two rows of staggered, corrosion resistant staples – carry out stapling on welted seams with two rows of staggered, corrosion resistant, staples.

17.0 Existing Underground Services

17.1 The Builder shall investigate the presence of existing underground services prior to commencement of any earthworks. Detection of underground services, which include electrical cables, water, sewerage and gas pipes, telecommunication cable, etc., must be determined and verified on site by an approved licensed service detection agency.

17.2 The Builder shall notify the S.O. immediately of any existing services such as water electrical and telephone cables underground drainage system sewers etc. which may be damaged by the works and shall immediately provide adequate supports and protection of same to the satisfaction of the S.O. at his own expense.

The approximate positions of some existing services known by the S.O. may be indicated on the Drawings. This is meant as a guide to the Builder and the accuracy thereof is not guaranteed. The Builder shall at his own initiative obtain up-to-date information from all the respective local authorities with regard to the actual positions of all existing services before commencement of earthworks. The Builder shall be liable and shall reinstate at his own expense any damage done during the course of the works.

If any existing services private or otherwise obstruct the works, the Builder shall notify the S.O. for his instruction. The S.O. shall at his sole discretion arrange for the resiting of the services if deemed necessary and the Employer shall bear the cost of such diversion. The Builder shall give every possible assistance and co- operation in the diversion re-alignment of existing services which are intended to be carried by the Builder or other local authorities.

18.0 Builder To Pay All Rates and Charges

The Builder is responsible for paying all rates and charges which may be levied by the local authorities in connection with temporary buildings or sanitary installations etc. erected or provided for within the site of the works for the purposes of this contract.

19.0 Interference With Traffic and Adjoining Properties

19.1 All operations necessary for the execution of the works shall so far as compliance with the requirements of the contract permits be carried on so as not to interfere unnecessarily or improperly with public convenience or the access to use and occupation of public or private roads and foot-paths and shall save harmless and indemnify the owner in respect of all claims demands proceedings damages costs charges and expenses whatsoever arising out of or in relation to any such matters.

19.2 The Builder shall where necessary or when required by the S.O. or by the competent relevant Authority apply and operate suitable methods of traffic control at his own expense.

20.0 Temporary Vehicular Crossings

The Builder shall be responsible for the design and construction of temporary vehicular crossings to the site. He shall engage a Professional Engineer to submit the necessary details and calculations to the relevant Authorities for approval prior to commencement of the works.

21.0 Road Openings

The Builder shall comply with LTA's Standard Conditions for openings within Road Reserves.

21.1 Application for Road Opening

The Builder shall submit through his Professional Engineer, all necessary applications in respect of road openings and obtain the required clearances/approvals from the relevant authorities prior to road opening for construction of external works which include sewer connections, drainage works and / or other underground services within existing road reserves.

21.2 Supervision

The Builder's Professional Engineer shall supervise the excavation and backfilling of all trenches and any reinstatement of road opening and submit his Certificate of Supervision to the Land Transport Authority (LTA) upon completion of the works.

21.3 Defects Liability

The Builder shall maintain any road reinstatement works for a period of one year and shall carry out from time to time, all necessary repair works deemed necessary by the LTA.

22.0 Turfing

Where turfing is required the area to be turfed must be levelled and made to slope at minimum 1/40 towards nearest drain and a minimum thickness of 25mm thick loamy top soil to be spread over the whole area before approved turf is to be planted. The turf must be maintained and cut throughout the maintenance period.

The types of grass acceptable or equivalent locally grown grass for transplanting are the following:

a) Axanophus (Cow Grass)
b) Digitaria (Serangoon)
c) Aechemum (Siglap)

The grass shall be of pure stock in tufts of 100mm length intact with root. The newly planted grass shall be well watered until survival is ensured.

Rolling shall be required if lumps and unevenness occurs.

The use of any type of grass shall first be approved by the S.O. and / or the local authorities prior to planting.

23.0 Post-Construction Survey

The Builder shall submit post-construction survey plan certified by a registered surveyor showing as-constructed levels at 10m grids, platforms, embankments, drainage, etc. of earthwork carried out.

24.0 Site/Pollution Control Measures

24.1 All construction activities shall comply with relevant sections of the latest editions of 'Code of Practice on Surface Water Drainage' and 'The Surface Water Drainage Regulations, and the requirements of PUB (Catchment and Waterways).

24.2 Earth Control Measures (ECM)

a) The Builder shall implement earth control measures (ECM) before any work commences. ECM carried out shall be in accordance with the requirements of the Catchment & Waterways Department of PUB, and the Builder shall engage a Qualified Erosion Control Professional (QECP) to submit and obtain approval of ECM proposals from the PUB.

b) The Builder shall also set up a ECM site management system to monitor the effectiveness of ECM on site, including daily site inspection reports by an Environmental Control Officer (ECO) / Supervisor engaged by the Builder. Such reports shall be endorsed by the QECP.

c) The Builder shall maintain the ECM until completion of all works, and shall provide a continuous monitoring system to ensure that the quality of any discharge into storm water drainage system is within the prescribed limits set by the PUB. Close circuit television (CCTV) camera/s shall also be provided to monitor the quality of discharge at outlets.

24.3 Pest and Vector Control

The Builder shall implement comprehensive pest control and surveillance for the site, including all necessary measures to prevent the site from being favourable to the breeding or harbouring of vectors. These shall be carried out in accordance to the requirements of the Environmental Health Department.

25.0 Verification and Submission

25.1 Submissions

25.1.1 Method of Statement

Before work commences, the Builder shall provide the S.O. with a method statement including at least the following information as relevant:

a) Proposed excavation and compaction plant
b) Maximum depth of each compacted layer
c) Minimum number of passes per layer
d) Method of treating slopes
e) Protection of formation and slopes
f) Proposed temporary works
g) Dewatering control

25.1.2 Programme of Works

Submit to the S.O. prior to the start of the Work unless otherwise agreed.

25.1.3 Addl. Site Investigation Factual Report

Extend to the S.O., a copy of the factual report for additional site investigation as applicable.

25.1.4 Temporary Works Design

a) General

Before commencement of Work, submit details of proposals for supporting the sides of all excavation which have slopes steeper than 1:2, or less if poor conditions exist, or exceeding 1.5m in depth which will be formed during the course of the Works. Such proposals shall be prepared by the Builder's Professional Engineer.

Provide adequate information to justify the adequacy of his proposals, if so requested.

b) Flow Chart for Ground Movement Monitoring

Submit for the S.O.'s acceptance, proposed ground movement instrumentation monitoring and critical levels.

25.1.5 Quality Control Plan

Prepare and submit the quality control plan S.O.'s acceptance prior to starting work.

25.2 Fill Material Sample

At least 21 days before filling work commences, the Builder shall submit full details of the sources and types of the proposed filling materials together with 25kg representative samples of each type from each source.

On request from the S.O. collect and dispose of these samples

At the same time, deliver to an accepted accredited laboratory, sufficient representative samples of each material for the specified tests, at a rate of one set of tests per sample.

25.3 Testing of Fill Materials

25.3.1 The Builder shall arrange for the following tests on all proposed fill materials at the rate of one test for each representative samples to be carried out at an accredited testing agency or equivalent local code requirements:

a) BS 1377-1 Section 3 : Classification
b) BS 5930 Site Investigation
c) BS 1377-2 : Determination of the particle size distribution
d) BS 1377-2 : Determination of the plasticity index (of material passing the 425 micron BS sieve; % of material retained to be reported)
e) BS 1377-3 : Determination of the total sulphate content
f) BS 812-111 : Determination of the ten per cent fines value.

The following tests may be required for certain materials as directed:

a) BS 1377-4 : Determination of the dry density / moisture content relationship of granular soil (vibrating hammer method)
b) BS 1377-4 : Determination of the California Bearing Ratio
c) BS 6906 : Part 8 : Determination of sand-geotextile frictional behaviour by direct shear
d) BS 5835 : Determination of strength of aggregates.

25.3.2 When directed, test all imported fill for compliance with specified limits on contamination to ensure that material is ‘clean’ relative to the proposed end use. Propose and submit the specific criteria, schedule and timing of tests required to S.O. for acceptance.

25.4 Cost of All Test

The Builder shall bear the costs of all construction tests and testing of materials.

26.0 Document Control

26.1 Approval: GLOBAL_FAC_GFTT_APPR

26.2 Notification:

GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONSTRUCTION
GLOBAL_FAC_ NOTIFY
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10_FAC_NOTIFY
FCG_F11_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY
FCG_MTB_FAC_NOTIFY

26.3 Notification:

All Micron Team Members at all Sites

26.4 Retention

Adherence to the requirements specified in
Global Facilities - Document Control Procedures

26.5 Review

Biennially (two years)

27.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New Document	07/05/2022